



**QUALITY MANAGEMENT PLAN
WAH4HOSPITAL (WAH4H)**

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MAKATI CITY, METRO MANILA 1232**

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INTRODUCTION

3.5.1.1 Document Purpose

This Quality Management Plan (QMP) establishes the quality framework governing all development, testing, documentation, and delivery activities for the WAH4H Hospital Information System project. It defines how quality is planned, assured, controlled, and measured throughout the project lifecycle — from schema design through formal system turnover to the WAH organization.

The QMP is a binding reference for Team Pseudoers and all WAH stakeholders involved in the project. It is aligned with the Project Charter (v1.0) and the Work Breakdown Structure (WBS), and must be read in conjunction with those documents. Any quality gate defined in this plan takes precedence over schedule pressure — quality acceptance criteria are non-negotiable conditions for milestone completion.

3.5.1.2 Project Background

Small municipal hospitals in the Philippines continue to operate under largely manual, paper-based clinical and administrative workflows. These environments are vulnerable to data entry errors, documentation delays, billing inconsistencies, and non-compliance with Philippine health informatics standards — all of which directly affect patient care quality and institutional eligibility for PhilHealth reimbursements.

WAH4H addresses this gap by delivering a fully digital, PH Core FHIR R4-compliant Hospital Information System that covers the complete clinical journey: from patient registration through admission, monitoring, discharge, pharmacy dispensing, laboratory testing, billing, and PhilHealth eClaims submission. The system is built for deployment in small municipal hospitals operating under the WAH (We are Hospital) organization and is designed to integrate with the broader WAH platform ecosystem — WAH4P (patient-facing) and WAH4PC (community/portal).

The current project deliverable is a demonstration version of WAH4H — a fully functional proof-of-concept intended for presentation to government officials (governors and mayors) to secure institutional buy-in and funding. A production-ready version will be developed in a subsequent engagement following successful demonstration. Post-delivery maintenance, feature enhancements, and operational support are explicitly out of scope for Team Pseudoers under the current engagement.

3.5.1.3 System Scope Summary

WAH4H covers 11 modules across two delivery phases:

- PBL1 (Core — initial UAT feedback received; UAT-requested changes in progress; final UAT/sign-off pending): Accounts, Patients, Admission, Monitoring, Discharge, Billing, Pharmacy, Laboratory



- PBL2 (Additional — Required in Final Build): Admin, Appointments, eClaims (PhilHealth)
- Cross-System: Data synchronization with WAH4P and WAH4PC platforms

The system is built on Django 5.x (backend), React 18 + Vite + TailwindCSS (frontend), and PostgreSQL (database), containerized via Docker and deployed on WAH-managed Proxmox infrastructure. Authentication is JWT-based. Access control enforces seven defined hospital roles: Doctor, Nurse, Pharmacist, Laboratory Technician, Billing Staff, Admin, and Hospital Admin.

3.5.1.4 Quality in Dimensions

Quality in this project is assessed across three interconnected dimensions, all of which must be satisfied before formal system turnover is initiated:

Dimension	Definition	Primary Responsibility
Standards Compliance	Full adherence to PH Core FHIR R4, ICD-10-PH, LOINC, SNOMED CT, MIMS PH, PSGC, RVS, NHFR, PhilHealth eClaims protocols, RA 10173, and RA 11223 across all applicable modules.	Jhon Lloyd Nicolas (Lead Developer / DevOps)
System Correctness	All 11 modules function as specified: correct RBAC enforcement, data integrity across module boundaries, accurate billing and eClaims computation, complete audit logging, and proper FHIR resource linkages.	Jhon Lloyd Nicolas / Mariyah Chavez
Deliverable Completeness	All project deliverables — three turnover manuals (Training, End-User, Technical), UAT documentation package, source code and repository — are produced, reviewed, and formally accepted by WAH before project closure.	Elijah Josh R. Quibin (PM)

3.5.1.5 Scope of This Document

This QMP covers the following:

- Quality management approach and governing principles
- All applicable regulatory, interoperability, and system quality requirements
- Quality Assurance (QA) — process-level activities that prevent defects
- Quality Control (QC) — testing and inspection activities that identify and resolve defects
- Quality metrics tracked throughout the project lifecycle
- Roles and responsibilities across the quality process (RACI)
- Change and continuous improvement procedures



This document does not cover project scheduling, cost management, or stakeholder communication in detail — those are addressed in the Project Charter, Schedule Baseline, and Communications Management Plan respectively.

QUALITY MANAGEMENT APPROACH

3.5.2.1 Philosophy

The quality approach for WAH4H is prevention-first. Defects discovered at the schema level cost significantly less to resolve than those discovered at UAT or, worse, post-deployment. This is especially critical given that WAH4H must comply with PH Core FHIR R4 — a standard that governs the structure of clinical data at the database field level. Retrofitting standards compliance after feature development creates compounding mismatches across module boundaries and requires extensive re-testing.

The team has already completed UAT across all PBL1 and PBL2 modules. The current phase involves post-UAT defect resolution, re-testing, documentation production, and formal system turnover. In this closing phase, quality focus shifts from defect prevention to defect elimination, documentation accuracy, and acceptance gate management.

The project operates under a formal three-tier deployment pipeline (Local → Sandbox → Production). No code or configuration change is promoted directly to the Production VM without first passing validation in Sandbox. Technical deployment and DevOps implementation are owned by Jhon Lloyd Nicolas, while Mariyah Vanna Monique C. Chavez validates QA evidence, environment parity, and handoff readiness.

3.5.2.2 Governing Principles

Principle	Description	Enforced By
Compliance Before Features	PH Core FHIR R4 schema alignment is established at the data layer before feature development begins on any module. No module build starts without a confirmed FHIR resource mapping and reference table population plan.	Jhon Lloyd Nicolas
Pipeline-Gated Promotion	All code must pass the three-tier pipeline: Local → Sandbox → Production. Sandbox must pass full integration and RBAC testing before Production VM deployment is approved by the PM.	Mariyah Chavez / PM
Role-Specific Testing	All 7 RBAC roles are tested independently across all 11 modules. Cross-role access	QA / Lead Developer



	violations are classified as Critical defects and block UAT sign-off.	
Traceable UAT	Every module has documented test cases tied to functional requirements from the WBS and Charter. Defects are logged in the UAT Defect Log with severity, resolution status, and re-test confirmation.	PM / Lead Developer
Formal Acceptance Gate	System turnover does not proceed until Ms. Anna Katrina Yturralde (Health Program Supervisor) formally approves all UAT outcomes, training delivery, and documentation completeness. Sir Pey Canlas provides final sign-off only after Ms. Anna Katrina Yturralde's approval — no bypass.	PM / Ms. Anna Katrina Yturralde / Sir Pey Canlas
Documentation Parity	All three turnover manuals (Training, End-User, Technical) must accurately reflect the final delivered system state. Documentation is reviewed against the live system before submission — not against earlier design drafts.	PM / All Members
Standards Confirmation Pending	Formal written confirmation from Sir Pey Canlas of the contractual compliance standards required by WAH is outstanding as of May 2026. Until confirmed, all standards listed in this plan are treated as required and implemented in full.	PM / Sir Pey Canlas

3.5.2.3 Quality in the Project Lifecycle

Quality activities are embedded across all WBS phases, not applied as a separate workstream at the end:

WBS Phase	Quality Activity
1.1 Project Management & Planning	QMP established. Acceptance criteria defined in Charter. Risk register captures quality-impacting risks (FHIR compliance gaps, UAT defects, interop spec delays).
1.2 System Design & Compliance Architecture	PH Core FHIR R4 Compliance Matrix produced before any module build. RA 10173 and RA 11223 compliance reviewed at architecture level.
1.3 Infrastructure Preparation	Three-tier pipeline validated. Docker environment parity verified across Local, Sandbox, and Production tiers. Cloudflare Tunnel and Fail2Ban configured.



1.4 & 1.5 Module Development (PBL1 + PBL2)	Schema reviewed per module before build. Code reviewed before merge. Module-level testing by developer before integration. RBAC enforced per role per module.
1.6 Cross-System Interoperability	WAH4P and WAH4PC sync tested end-to-end. Cross-platform FHIR resource references validated.
1.7 Security Implementation	RBAC matrix tested for all 77 role-module combinations. JWT expiry/refresh tested. Audit log coverage verified per write operation.
1.8 UAT & Defect Resolution	Formal UAT executed with WAH stakeholders. All defects logged, resolved, and re-tested. Sign-off from Anna Katrina Yturralde required before proceeding to turnover.
1.9 Turnover Documentation	All three manuals reviewed for accuracy against live system. UAT documentation package compiled. Source code and deployment scripts verified.
1.10 System Deployment & Handoff	Production VM deployment validated. Sandbox-to-Production parity verified. WAH staff training delivered. Formal handoff executed per sign-off chain.
1.11 Project Closure	Project Closeout Report includes quality summary: UAT outcomes, defect resolution totals, compliance verification status, and lessons learned.

QUALITY REQUIREMENTS / STANDARDS

3.5.3.1 Overview

WAH4H operates under a comprehensive set of regulatory, interoperability, and functional quality requirements. These requirements are derived from the Project Charter (Section 5.3 and Section 6), the Compiled Interview Notes (Sir Pey Canlas and Sir Robert Michael Martinez), and the WBS (Sections 1.2, 1.4, 1.5, and 1.7). Full compliance across all standards listed below is required before Ms. Anna Katrina Yturralde (Health Program Supervisor) can issue UAT sign-off and before Sir Pey Canlas can provide final project acceptance.

3.5.3.2 Regulatory and Interoperability Standards

Standard	Application in WAH4H	Affected Modules	Verification Method
PH Core FHIR R4	All clinical resources (Patient, Encounter, Observation, Condition, Procedure, MedicationRequest, DiagnosticReport, Claim, ClaimResponse) must conform to PH Core profiles. Compliance	All 11 modules	FHIR R4 Compliance Matrix + schema review per module + FHIR validator output



	established at schema level before feature development.		
NHFR	WAH facility must be registered in the National Health Facility Registry. Required for Location resource registration and eClaims compliance. NHFR registration is a WAH prerequisite — outside Team Pseudoers scope.	Admin, eClaims	WAH confirmation of NHFR registration before eClaims go-live
PSGC	Philippine Standard Geographic Code applied to all address fields — patient registration (region, province, city/municipality, barangay) and facility address in Admin module.	Patients, Admin	PSGC-coded dropdown enforcement; code presence validated on form submission
PhilHealth eClaims	CF1, CF2, and CF4 form generation per DOH/PhilHealth specifications. Batch submission compliant with PhilHealth eClaims system. WAH PhilHealth accreditation is a prerequisite outside project scope.	eClaims, Billing	Form output review against CF specs; batch submission test with PhilHealth system
ICD-10-PH	Philippine adaptation of ICD-10 applied to all diagnosis fields: admitting diagnosis (Admission), active diagnoses (Monitoring), principal and discharge diagnoses (Discharge), and claim diagnoses (eClaims).	Admission, Monitoring, Discharge, eClaims	ICD-10-PH code lookup enforcement on input; reference table completeness audit
LOINC	Logical Observation Identifiers Names and Codes applied to laboratory test identifiers and clinical observation codes in Monitoring.	Laboratory, Monitoring	LOINC reference table constraint; code validated on test/observation creation
SNOMED CT	Systematized Nomenclature of Medicine for clinical findings, conditions, and procedures in Monitoring and Discharge modules.	Monitoring, Discharge	SNOMED CT reference table constraint; validated on clinical record creation



MIMS PH	Philippine drug reference applied to all medication records. All dispensed drugs reference MIMS PH drug codes in Pharmacy module.	Pharmacy	MIMS PH drug lookup constraint on prescription processing and dispensing
RVS	Relative Value Scale procedure coding applied to PhilHealth claim procedures. All billed procedures use RVS codes.	eClaims, Billing	RVS code lookup enforcement on procedure entry; validated in claim generation output
RA 10173 (Data Privacy Act)	All patient data encrypted at rest and in transit. Access to patient records logged with user ID, timestamp, and action. Audit logs tamper-evident and accessible to Hospital Admin role only.	All modules	Security audit; encryption verification; audit log coverage test per write operation
RA 11223 (UHC Act)	PhilHealth interoperability and Universal Health Care alignment for eClaims data exchange and benefit processing.	eClaims, Billing	Functional test of eClaims submission and remittance advice processing

3.5.3.3 Functional Quality Requirements

The following functional quality requirements define the minimum acceptable behavior of the delivered system per module. Acceptance criteria are specific and measurable — each criterion must be verifiable during UAT execution and confirmed by Ms. Anna Katrina Yturalde (Health Program Supervisor) before module sign-off is granted. All 11 modules must achieve their acceptance criteria prior to formal project turnover.

Module	Specific Requirement	SMART Acceptance Criterion	Verif By
Accounts	RBAC enforced for all 7 roles (Doctor, Nurse, Pharmacist, Lab Tech, Billing Staff, Admin, Hospital Admin) across all 11 modules. JWT access tokens expire after configured duration; refresh tokens function correctly. Role dashboards surface only role-permitted actions and data.	0 of 77 role-module test combinations result in unauthorized data access or action execution at both UI and API layer. 100% of DRF API endpoints return HTTP 401 when tested with an expired or absent Bearer token. Token refresh endpoint returns a valid new access	Jhon Lloyd Nicol Mariy Chav



		token within 2 seconds of a valid refresh token submission.	
Patients	Patient registration form enforces 9 mandatory fields: PhilHealth ID, last name, first name, date of birth, sex, PSGC-coded complete address (region, province, city/municipality, barangay), blood type, civil status, and at least 1 emergency contact. PH Core FHIR R4 Patient resource generated on save.	Form submission with any of the 9 mandatory fields empty is rejected with a field-level validation error — tested across 9 separate negative test cases. PSGC codes validated against the PSGC reference table — free-text address entry is blocked. FHIR Patient resource output passes HL7 FHIR validator against the PH Core Patient profile with 0 errors (warnings acceptable).	Jhon Lloyd Nicol
Admission	Admission record captures: admission type, attending physician (linked to PractitionerRole), ward/room assignment (linked to Location resource), ICD-10-PH admitting diagnosis code, and PhilHealth/HMO insurance details. FHIR Encounter resource created on admission save. Encounter references the Patient resource and links to Billing.	FHIR Encounter resource created on every admission save — verified by querying the Encounter endpoint for the created record. ICD-10-PH code field rejects free-text input and requires selection from ICD-10-PH reference table — tested with 3 invalid free-text entries. Encounter.subject references the correct Patient resource ID. Billing record is created and linked to the Encounter ID — verified via Billing module lookup by Encounter ID.	Jhon Lloyd Nicol
Monitoring	All 5 vital sign types (Temperature [deg C], Systolic/Diastolic BP [mmHg], Pulse Rate [bpm], Respiratory Rate [breaths/min], O2 Saturation [%]) recorded as individual FHIR Observation resources with correct LOINC codes. Nursing notes and physician orders saved with author, timestamp,	Each of the 5 vital sign observation types maps to its correct LOINC code: Temperature=8310-5, BP Systolic=8480-6, BP Diastolic=8462-4, Pulse Rate=8867-4, O2 Sat=2708-6,	Jhon Lloyd Nicol



	and linked Encounter. Intake/output entries tracked with volume (mL) and category.	RR=9279-1 — verified in FHIR Observation.code.coding for each type. Clinical range validation rejects: Temperature outside 30-45 deg C, O2 Sat outside 50-100%, PR outside 20-250 bpm — tested with out-of-range inputs for each.	
Discharge	Discharge record captures: ICD-10-PH principal discharge diagnosis, discharge type (Recovered/Improved/HAMA/Transferred/Expired), discharge summary text, and follow-up instructions. FHIR Encounter.hospitalization.dischargeDisposition updated. Discharge clearance document generated as a downloadable output containing patient name, admission date, discharge date, attending physician, and discharge diagnosis.	FHIR Encounter.status updated to "finished" and dischargeDisposition populated on discharge save — verified via Encounter API response. Discharge clearance document generated within 5 seconds of discharge confirmation — tested in 5 consecutive discharge scenarios. Document contains all 5 required fields (name, admission date, discharge date, physician, discharge diagnosis). ICD-10-PH code required — form rejects save without a valid code.	Jhon Lloyd Nicol
Billing	Billing module generates itemized invoice per admitted patient listing each charged service line (room, procedure, medication, lab). PhilHealth benefit deduction computed per applicable benefit package. HMO deduction applied where applicable. Patient payable = Total Charges minus PhilHealth Deduction minus HMO Deduction. Official Receipt (OR) generated on payment confirmation.	Billing computation tested against 5 predefined test cases with known charge inputs and expected patient payable outputs — all 5 must return the correct computed amount with 0 peso variance. PhilHealth deduction field present and populated on all records flagged as PhilHealth-covered. OR document generated within 5 seconds of payment confirmation — tested in 5 scenarios. OR	Jhon Lloyd Nicol



		contains: OR number, patient name, amount paid, payment date, cashier name — all 5 fields verified per OR output. Note: Billing PHCore FHIR alignment is the highest-risk compliance gap — FHIR Invoice and ChargeItem resource conformance must be validated against PH Core profiles before UAT sign-off.	
Pharmacy	Medication dispensing linked to a FHIR MedicationRequest from the attending physician. All dispensed items reference a MIMS PH drug code. Drug inventory stock quantity decremented by dispensed quantity on each dispense confirmation. Low-stock alert triggered when any drug's current stock falls at or below the configured minimum threshold.	MIMS PH code required on every dispense record — form blocks save without MIMS PH code selection, tested with 3 attempted free-text entries. Stock quantity for a test drug decremented by the exact dispensed quantity after each of 5 consecutive dispense operations. Low-stock alert appears in the Pharmacy dashboard within 1 page refresh of stock falling to or below threshold. FHIR MedicationRequest.subject references correct Patient and MedicationRequest.status updated to "completed" on dispensing.	Jhon Lloyd Nicol
Laboratory	Lab test requests linked to ordering physician (PractitionerRole) and patient (Patient resource). All test types use LOINC codes from the lab test reference table. Specimen tracking records specimen type, collection date/time, and collector. Results entered per test item. Result release action notifies attending physician's record and updates FHIR DiagnosticReport.status to "final".	LOINC code present on 100% of test requests — verified by querying all test requests created during UAT and checking DiagnosticReport.code.coding. FHIR DiagnosticReport.status = "final" after result release — verified via API for each released result in UAT. Result	Jhon Lloyd Nicol



		release action creates a notification entry linked to the attending physician's PractitionerRole ID. Specimen collection date/time field rejects future timestamps — tested with a future date input.	
Admin	Facility record maps to FHIR Organization resource. Department/ward records map to FHIR Location resources with Location.managingOrganization referencing the facility Organization. Staff records map to FHIR PractitionerRole with linked Practitioner and Organization. Audit logs capture all system write operations with user ID, timestamp, action, and affected record ID. Audit log view accessible only to Hospital Admin role.	FHIR Organization resource created for WAH facility — validated with 0 errors against PH Core Organization profile. FHIR Location created for each department/ward with correct managingOrganization reference — tested for 3 department records. Audit log view returns HTTP 403 for all 6 non-Hospital-Admin roles — tested for each role. Audit log entry present for 100% of write operations performed during a 10-minute monitored test session.	Jhon Lloyd Nicol
Appointments	Appointment booking workflow creates FHIR Appointment resource with status "booked", linked to Patient and PractitionerRole. Rescheduling updates Appointment.start and Appointment.end. Cancellation updates Appointment.status to "cancelled". Calendar view displays all booked appointments for a given physician or department filtered by date range.	FHIR Appointment resource created on booking with status="booked", correct Patient reference, and correct PractitionerRole reference — verified via Appointment API for 5 booking test cases. Rescheduled appointment reflects updated start/end times in FHIR Appointment and in calendar view within 1 page load. Calendar view for a given physician displays only that physician's appointments. Cancellation updates Appointment.status	Jhon Lloyd Nicol



		to "cancelled" in FHIR — verified via API query after cancellation action.	
eClaims	CF1, CF2, and CF4 form outputs contain all fields mandated by PhilHealth eClaims specifications. All procedure items in a claim carry an RVS procedure code. Batch submission file generated in the format accepted by the PhilHealth eClaims system. Remittance advice import updates claim status and records the approved amount per claim. Payment reconciliation dashboard shows total submitted, total approved, total rejected, and net receivable amounts.	CF1/CF2/CF4 form field coverage verified against the PhilHealth eClaims field specification checklist — 0 mandatory fields missing across 3 test claim records. RVS code present on 100% of procedure items in all generated claims — verified by inspecting claim output for 5 test cases. Remittance advice import correctly updates claim status and approved amount for all claims in a test remittance file with at least 5 claim entries. Payment reconciliation totals match manual sum of test claim records — 0 peso variance across all 4 reconciliation fields.	Jhon Lloyd Nicol
WAH4P / WAH4PC Interop	Patient and clinical data synchronized with WAH4P and WAH4PC platforms per sync events to be formally defined by Sir Robert Michael Martinez (open item: Interview Notes Section 9, Item 2). FHIR resource references maintained across platform boundaries.	100% of defined sync event types trigger a sync call within 30 seconds of the triggering action in WAH4H — verified in integration test environment. Data fields in WAH4P/WAH4PC target records match source WAH4H records field-for-field with 0 mismatches across 10 test sync events per event type. Cross-platform FHIR resource references resolve correctly in the target system — verified for each linked resource type.	Jhon Lloyd Sir Robe Mich Mart



3.5.3.4 Non-Functional Quality Requirements

Category	Specific Requirement	SMART Acceptance Threshold	Test Method
Security — RBAC	Each of the 7 roles enforced independently across all 11 modules. Unauthorized actions blocked at both API (Django REST Framework permission classes) and UI (conditional rendering) layers.	0 unauthorized data accesses or action executions out of 77 role-module test combinations. Tests performed by logging in as each role and attempting all restricted actions — both via UI and direct API calls with valid JWT for that role.	Manual test matrix: 7 roles × 11 modules executed by Jhon Lloyd Nicolas and Mariyah Chavez. Results recorded in RBAC Test Matrix spreadsheet.
Security — JWT Auth	JWT access tokens expire after the configured duration (e.g., 60 minutes for access token, 7 days for refresh token). All DRF API endpoints decorated with <code>IsAuthenticated</code> or equivalent. Refresh endpoint issues a new valid access token on valid refresh token submission.	100% of API endpoints return HTTP 401 when called with an expired token. 100% of endpoints return HTTP 401 with no token. Refresh endpoint returns HTTP 200 and a new access token within 2 seconds for a valid refresh token — tested 5 times consecutively.	Automated test script hitting all API endpoints with expired/absent tokens. Refresh tested manually and via script. Results logged per endpoint.
Security — Encryption	All patient-identifiable fields in PostgreSQL encrypted at rest using Django field-level encryption or equivalent. All external traffic routed through Cloudflare Tunnel (HTTPS/TLS 1.2+). No plaintext patient data in application logs, error responses, or API responses outside authenticated sessions.	0 plaintext patient records found in PostgreSQL after querying the patients table directly from psql (bypassing the application layer). Cloudflare Tunnel active — direct port access to the server returns connection refused on all	Mariyah Chavez, direct DB query + port scan + log review



		application ports. 0 patient data fields visible in application error logs or debug output — verified by triggering 5 intentional errors.	
Audit Logging	Every POST, PUT, PATCH, and DELETE operation across all 11 modules generates an audit log entry containing: authenticated user ID, timestamp (UTC), HTTP method, affected resource type, affected record ID, and HTTP response status.	Audit log entry count equals the total number of write operations performed during a structured 30-minute test session — 0 missed log entries. Audit log entries for all 11 modules verified by performing at least 2 write operations per module and confirming 22 audit entries are present. Audit log view returns HTTP 403 for all 6 non-Hospital-Admin roles — tested for each role.	Jhon Lloyd Nicolas — structured test session with counted writes vs. log entries. Role access tested by Mariyah Chavez.
Infrastructure Security	Cloudflare Tunnel configured as the sole entry point for external traffic. Fail2Ban active on Proxmox host with ban threshold set to 5 failed login attempts within 60 seconds. No application ports (8000, 5432, etc.) exposed to public internet.	Port scan (nmap) of the Proxmox server's public IP returns 0 open application ports. Fail2Ban bans a test IP after exactly 5 failed attempts within 60 seconds — verified by monitoring /var/log/fail2ban.log. Cloudflare Tunnel operational — WAH4H accessible via tunnel URL and inaccessible via direct server IP.	Mariyah Chavez — nmap scan + Fail2Ban test + tunnel access verification. Executed and documented before Production handoff.



Deployment Pipeline	Three-tier pipeline enforced: Local → Sandbox → Production. Sandbox promotion requires passing the full test suite. Production promotion requires written PM approval (Elijah Quibin) documented in the pipeline log.	0 code changes deployed directly to Production without a corresponding Sandbox validation entry — verified in the deployment log at turnover. All Production releases reference a Sandbox validation timestamp and PM approval entry in the pipeline log.	Mariyah Chavez — pipeline log reviewed at each deployment event. PM countersigns the log entry for each Sandbox-to-Production promotion.
Data Integrity	No orphaned records in any module. All FK constraints enforced at the PostgreSQL level. FHIR resource references (Encounter→Patient, Observation→Encounter, DiagnosticReport→Encounter, Invoice→Patient) valid and resolvable.	0 orphaned records found by running referential integrity check queries across all 11 module tables before UAT sign-off. 0 FK constraint violations in the final database state. FHIR validator returns 0 errors on all FHIR resources generated during UAT test session.	Jhon Lloyd Nicolas — SQL integrity check scripts run against Sandbox DB. FHIR validator run against exported resource samples per module.
Performance	All standard module API endpoints (list, retrieve, create) respond within acceptable time under normal single-user load — defined as a single concurrent user executing standard module operations.	100% of tested API endpoints return HTTP 200 responses in under 3 seconds during manual load testing performed by a single tester — measured using browser dev tools network tab for at least 3 requests per endpoint. Patient search returning up to 100 records responds in under 3 seconds.	Jhon Lloyd Nicolas — manual timing test using browser dev tools. Results logged per endpoint in performance test log.



QUALITY ASSURANCE

3.5.4.1 Overview

Quality Assurance in WAH4H encompasses all process-level activities designed to prevent defects from entering the codebase, database schema, or project deliverables. QA is proactive — it operates ahead of testing to ensure that the right processes are followed, the right standards are applied, and the right gates are respected throughout the project lifecycle.

QA is the primary responsibility of Mariyah Vanna Monique C. Chavez (QA), working with Jhon Lloyd Nicolas (Lead Developer / DevOps) for technical fixes and deployment-related implementation. The Project Manager (Elijah Josh R. Quibin) is accountable for ensuring QA activities are completed at each milestone before advancing to the next phase. Sir Robert Michael Martinez (Technical Support Supervisor, WAH) is consulted on QA activities related to infrastructure and interoperability.

Given that WAH4H is currently in the post-UAT phase (defect resolution, documentation, and turnover preparation), QA focus is on: (1) ensuring all post-UAT improvements are fully re-tested before turnover; (2) ensuring all documentation accurately reflects the current system state; and (3) ensuring the Sandbox-to-Production VM deployment is validated before handoff.

3.5.4.2 QA Activities

Activity	Description	Trigger / Frequency	Owner
FHIR Schema Review	Before each module build, the database schema is validated against PH Core FHIR R4 profiles. All fields must have an explicit standard mapping or a documented rationale for deviation. No module development starts without a completed schema review.	Before each module build	Jhon Lloyd Nicolas
Reference Table Audit	All coding reference tables — ICD-10-PH, LOINC, SNOMED CT, MIMS PH, PSGC, and RVS — are audited for completeness and correct code values before the corresponding clinical module goes live. Incomplete reference tables are a blocking issue.	Before module launch (per module)	Jhon Lloyd Nicolas
Code Review	All code changes are reviewed by the Lead Developer / DevOps before	Every pull request / merge	Jhon Lloyd Nicolas



	merge to the main branch. Review checks: FHIR compliance in resource construction, RBAC logic correctness, input validation, audit log trigger presence, and absence of hardcoded credentials or patient data.		
RBAC Matrix Validation	The RBAC access control matrix (7 roles × 11 modules = 77 combinations) is reviewed and updated after any module change. All access rules are tested to verify enforcement at both the UI and API levels.	Post-module change; pre-UAT	Jhon Lloyd Nicolas / Mariyah Chavez
Pipeline Gate Validation	Before any promotion between tiers: (1) Local → Sandbox: automated test suite passes; (2) Sandbox → Production: full integration test passes and PM grants promotion approval. Pipeline promotion logs maintained.	Per deployment event	Mariyah Chavez
Security Configuration Review	JWT configuration (expiry, refresh), Cloudflare Tunnel settings, Fail2Ban rules, and environment variable management reviewed at the end of each sprint. Secrets confirmed absent from source code repository.	Sprint end	Mariyah Chavez
Interoperability Spec Review	WAH4P and WAH4PC integration technical specification (to be defined by Sir Robert Michael Martinez) reviewed before integration sprint begins. Sync events, data fields, triggers, and FHIR cross-platform reference patterns confirmed before implementation.	Before integration sprint	Jhon Lloyd Nicolas / Sir Robert Michael Martinez
Post-UAT Improvement Review	After post-UAT defect resolution, all fixed items are reviewed against the original UAT Defect Log to confirm completeness. No module is considered complete until all critical and high defects are resolved and re-tested.	After each post-UAT fix cycle	PM / Lead Developer



Documentation QC Review	All three turnover manuals (Training, End-User, Technical) are reviewed against the live system before submission. Reviews check: accuracy of workflows, correctness of role-specific instructions, completeness of Proxmox/Docker setup steps, and validity of backup/restore procedures.	Before documentation submission	Elijah Josh R. Quibin
Deployment Parity Check	Before Production go-live, Sandbox and Production environments are compared for configuration parity: Docker Compose versions, environment variables, database schema version, seed data, and Cloudflare Tunnel routing.	Before Production VM deployment	Mariyah Chavez

3.5.4.3 QA Roles and Responsibilities

Role	Person	QA Responsibility
Project Manager	Elijah Josh R. Quibin	Accountable for QMP compliance across the project. Ensures QA gates are completed before milestone advancement. Owns documentation QC. Escalates unresolved QA issues to Sir Pey Canlas or Sir Robert Michael Martinez. Signs off on post-UAT readiness for turnover.
Lead Developer / DevOps	Jhon Lloyd Nicolas	Technical QA authority. Owns FHIR R4 schema compliance, RBAC enforcement logic, reference table integrity, code review, and module-level integration quality. Responsible for ensuring all PH Core profiles are applied correctly at the data layer before feature coding begins.
QA	Mariyah Vanna Monique C. Chavez	Owns infrastructure QA: pipeline validation, Docker environment parity, Proxmox configuration, Cloudflare Tunnel and Fail2Ban setup. Validates Sandbox-to-Production parity before handoff. Co-reviews RBAC enforcement at infrastructure level.
Scrum Master / Agile Developer	John Kenneth L. Jajurie	Ensures Agile ceremonies (sprint planning, retrospectives, reviews) surface quality risks and impediments. Tracks QA action items in sprint



		backlog. Facilitates retrospective-driven process improvements.
WAH Technical Supervisor	Sir Robert Michael Martinez	Technical acceptance for infrastructure deployment and interoperability. Reviews WAH4P/WAH4PC integration outputs. Confirms server-side deployment quality on Proxmox. Point of contact for WAH-side security requirements.
Acceptance Authority	Ms. Anna Katrina Yturalde (Health Program Supervisor)	Primary UAT sign-off authority. Reviews all module functionality against operational hospital workflows. Confirms training delivery is satisfactory. Must provide formal approval of all deliverables before Sir Pey Canlas issues final sign-off.
Project Sponsor	Sir Pey Canlas	Ultimate quality authority. Final sign-off after Ms. Anna Katrina Yturalde's approval. Must confirm contractual compliance standards required by WAH (open item — pending confirmation as of May 2026).

QUALITY CONTROL

3.5.5.1 Overview

Quality Control (QC) in WAH4H covers all inspection, testing, and verification activities that detect defects, validate outputs against acceptance criteria, and confirm that deliverables meet the standards defined in Section 3. While QA prevents defects from entering the process, QC identifies defects that exist and ensures they are resolved before any gate is passed.

The primary formal QC gate is the User Acceptance Testing (UAT) cycle. Initial UAT feedback has been received; UAT-requested changes are in progress; final UAT/sign-off is pending after re-testing. The current focus is on post-UAT defect resolution, re-testing, and pre-turnover QC activities including documentation review and Production VM deployment validation.

QC activities are conducted at four levels: (1) module level, (2) integration level, (3) system level (UAT), and (4) deployment level. Each level has defined pass criteria and documented actions for failures.

3.5.5.2 QC Process Flow

Stage	QC Activity	Pass Criteria	Failure Action
Module Build	Developer testing of module functionality	No critical functional gaps. FHIR resources	Fix before sprint demo. Defect logged. Not



	before sprint demo. FHIR resource output verified. RBAC rules tested for the module.	pass validator. RBAC enforced per role.	promoted to integration test until resolved.
Integration Testing	Module-to-module data flow tested: Admission → Monitoring, Monitoring → Discharge, Billing ← Admission/Pharmacy/Lab, eClaims ← Billing/Discharge.	Data passes correctly across all linked module boundaries. No orphaned records. FHIR resource references resolve.	Defect logged with affected module boundary. Fix assigned to current sprint. Integration test repeated after fix.
Security Testing	RBAC matrix tested for all 77 role-module combinations at UI and API layer. JWT expiry and refresh tested. Audit log coverage verified for all write operations.	Zero unauthorized access events. All expired tokens rejected (HTTP 401). Audit log entries present for 100% of write operations.	RBAC or JWT failures classified as Critical — block UAT entry. Audit log gaps classified as High. All must be resolved and re-tested before UAT.
UAT Execution	WAH stakeholders (led by Anna Katrina Yturralde) execute documented test cases across all 11 modules. Defects logged in UAT Defect Log with severity classification.	All high-priority test cases pass. Zero critical defects open at sign-off. Defect count within acceptable tolerance per module.	Critical and High defects logged in Defect Log. Fix cycle initiated. Re-test required before UAT sign-off. Ms. Anna Katrina Yturralde's sign-off withheld until all Critical/High resolved.
Post-UAT Defect Resolution	All critical and high defects from UAT fixed and re-tested. Fix confirmed by original tester where possible.	Zero open Critical defects. All High defects resolved or formally accepted with documented rationale.	Outstanding Critical defects prevent turnover. High defect deferrals must be approved by PM and communicated to Sir Robert Michael Martinez / Anna Katrina Yturralde.
Documentation Review	PM reviews all three manuals against live system state. Checks: workflow accuracy, role-specific instruction correctness, technical procedure validity	No inaccurate or missing content for any module, workflow, or role. Technical procedures verified executable.	Identified inaccuracies returned to author for correction. Document not submitted until PM review passes.



	(Proxmox, Docker, backup/restore).		
Production VM Deployment Validation	Post-deployment check: all containers running, database schema current, application accessible, RBAC enforced, audit logging active, Cloudflare Tunnel live.	Production VM behaves identically to Sandbox. Zero environment-specific defects at handoff.	Rollback to last Sandbox state. Identify and fix environment delta. Re-deploy and re-validate before formal handoff.
Formal Acceptance	Anna Katrina Yturralde reviews system and documentation completeness. Provides written sign-off. Sir Pey Canlas provides final written sign-off after Ms. Anna Katrina Yturralde's approval.	Written acceptance received from Anna Katrina Yturralde and Sir Pey Canlas per sign-off chain documented in Project Charter.	Issues raised by Anna Katrina Yturralde or Sir Pey Canlas re-enter defect resolution cycle. Turnover not closed until both sign-offs received.

3.5.5.3 Defect Classification and Management

Severity	Definition	Examples in WAH4H	Resolution Requirement
Critical	System crash, data loss, security violation, RBAC bypass, FHIR resource structural failure, eClaims form non-compliance.	Unauthorized role accessing patient billing records; Encounter resource fails PH Core validator; CF1 form missing required PhilHealth fields; audit logging completely absent on a module.	Must be resolved and re-tested before UAT sign-off. Blocks Production VM deployment. PM must be notified immediately on identification.
High	Major functional failure in a module workflow that prevents the workflow from completing, or significant data accuracy error.	Billing computation producing incorrect patient payable amount; eClaims batch submission failing; Pharmacy stock not decrementing after dispensing; discharge clearance not generating.	Must be resolved before final turnover. Re-test by original tester required. Outstanding High defects at sign-off must be formally accepted by Anna Katrina Yturralde with PM approval.
Medium	Non-blocking functional issue.	Non-critical UI field validation gap; minor	Resolved within the sprint it is raised. If deferred



	Workaround exists. Does not prevent module workflow completion.	display error on dashboard; PSGC dropdown not pre-filtering by region on first load; appointment notification delayed.	beyond sprint, formally documented with target date and approved by PM.
Low	Cosmetic or minor UX issue. No functional impact. Typos in documentation.	Button misalignment; inconsistent date format display; minor wording error in module label; documentation typo.	Resolved before documentation submission. Does not block any gate. Logged and batched for resolution.

3.5.5.4 UAT Documentation Requirements

Initial UAT feedback has been received across PBL1 and PBL2 modules; UAT-requested changes and re-testing are still in progress. The following documentation must be finalized and included in the turnover package (currently pending — assigned to Jhon Lloyd Nicolas, per Interview Notes Section 9, Item 7):

- UAT Test Cases: Documented test cases per module covering all functional requirements from the WBS, including role-specific scenarios for all 7 RBAC roles.
- UAT Test Results: Pass/fail outcomes per test case, with tester name, date executed, and actual vs. expected result recorded.
- UAT Defect Log: All defects identified during UAT with severity, description, assigned resolver, resolution date, and re-test confirmation.
- UAT Sign-Off Sheets: Formal sign-off sheet per module to document final UAT/sign-off once UAT-requested changes are implemented, re-tested, and accepted by Ms. Anna Katrina Yturralde (Health Program Supervisor).

UAT documentation is a non-negotiable deliverable for project closure, as confirmed by Sir Pey Canlas during the sponsor interview (Interview Notes, Section 3). Absence of complete UAT documentation will prevent formal project sign-off.

QUALITY CONTROL MEASUREMENTS

3.5.6.1 Overview

Quality metrics provide objective, quantifiable evidence of quality status throughout the project lifecycle. Metrics are reviewed at each sprint review, reported in the Milestone Tracking Record, and included in the Project Closeout Report. Metrics that fall below target thresholds trigger a quality escalation review with the PM and, where applicable, with Sir Robert Michael Martinez or Anna Katrina Yturralde.



Metrics are organized into four categories: (1) compliance metrics that track standards adherence, (2) testing metrics that track UAT and defect outcomes, (3) security metrics that track RBAC and audit coverage, and (4) delivery metrics that track documentation and deployment quality.

3.5.6.1 Quality Metrics

#	Metric	Specific Definition	SMART Target	Measurement Method & Owner
QM-01	FHIR Resource Compliance Rate	Count of FHIR resource types generated by WAH4H that pass the HL7 FHIR validator against PH Core FHIR R4 profiles with 0 errors, divided by total resource types implemented.	14/14 resource types (100%) pass PH Core FHIR R4 validation with 0 errors by the date the first UAT session is held with Anna Katrina Yturralde. Warnings are acceptable; errors are not. Validated per module as each is built.	Jhon Lloyd Nicolas runs FHIR validator against one sample resource per type per module. Results logged in the FHIR R4 Compliance Matrix with validator output attached. Reviewed by PM before UAT entry.
QM-02	Reference Table Completeness	For each of the 6 required coding reference tables (ICD-10-PH, LOINC, SNOMED CT, MIMS PH, PSGC, RVS), the percentage of codes required by the corresponding module that are present and correctly mapped in the WAH4H database reference table before that	All 6 reference tables are 100% populated with the required code sets before the module that depends on each table enters UAT. PSGC table covers all Philippine regions, provinces, and municipalities (1,634+ entries per PSGC 2023 publication).	Jhon Lloyd Nicolas runs a row count and sample spot-check query on each reference table. Results documented in the Reference Table Audit Checklist with table name, row count, sample entries verified, and sign-off date.



		module enters UAT.		
QM-03	UAT First-Pass Rate	Per module: number of UAT test cases that pass on the first execution attempt divided by total test cases for that module, expressed as a percentage.	Each of the 11 modules achieves $\geq 90\%$ first-pass rate at the UAT session. Modules with a first-pass rate below 90% re-enter defect resolution. Overall project first-pass rate (all modules combined) $\geq 92\%$.	PM (Elijah Quibin) tallies pass/fail per test case in the UAT Test Results Log during each UAT session with Anna Katrina Yturralde. Pass rate computed per module and overall after each session.
QM-04	Critical Open Defects at UAT Sign-Off	Number of defects classified as Critical severity that remain unresolved at the moment Anna Katrina Yturralde is requested to sign off on UAT.	0 Critical defects open at the time of UAT sign-off request. If any Critical defect is open, UAT sign-off is not requested until all Critical defects reach "Resolved and Re-tested" status in the Defect Log.	Jhon Lloyd Nicolas maintains the UAT Defect Log. PM verifies Critical defect count = 0 before scheduling sign-off meeting with Anna Katrina Yturralde.
QM-05	High Defects Open at UAT Sign-Off	Number of defects classified as High severity that remain in "Open" or "In Progress" status in the UAT Defect Log at the time of UAT sign-off submission.	0 unaccepted High defects at sign-off. If any High defect cannot be resolved before the sign-off meeting, PM must obtain Ms. Anna Katrina Yturralde's written acceptance of each deferred High defect by name, with a documented fix	PM tracks High defect status in UAT Defect Log. Ms. Anna Katrina Yturralde's written acceptance of any deferred High defect recorded as an attachment to the sign-off sheet.



			target date, before sign-off is granted.	
QM-06	Defect Resolution Velocity	Number of defects (any severity) raised in a given sprint that are resolved — confirmed fix plus successful re-test — within that same sprint's end date, divided by total defects raised in that sprint.	$\geq 85\%$ of defects raised in each sprint are resolved within the same sprint. If velocity drops below 85% in any sprint, PM escalates to Jhon Lloyd Nicolas within 1 business day and a resolution plan is produced before the next sprint starts.	PM cross-references sprint backlog and UAT Defect Log at each sprint review. Velocity computed per sprint and tracked in the Milestone Tracking Record.
QM-07	RBAC Test Coverage	Number of role-module combinations (7 roles $\times$ 11 modules = 77 total) for which a documented test case exists, has been executed, and has a recorded result (Pass or Fail) in the RBAC Test Matrix, at both UI and API layer.	77/77 combinations (100%) documented, executed, and recorded in the RBAC Test Matrix before UAT sign-off is requested. 0 untested combinations permitted. Matrix must be complete no later than 3 business days before the scheduled UAT sign-off meeting.	Jhon Lloyd Nicolas and Mariyah Chavez jointly execute and record the RBAC matrix. PM confirms 77/77 completeness before scheduling sign-off meeting with Anna Katrina Yturralde.
QM-08	Audit Log Write Coverage	Number of distinct write operation types (POST, PUT, PATCH, DELETE) per module for which an audit	100% of write operation types across all 11 modules generate a compliant audit log entry — verified by a structured test	Jhon Lloyd Nicolas performs write operations per module and counts audit log entries. Mariyah Chavez tests role-based access restriction. Results



		log entry is confirmed present and correctly populated, divided by total write operation types across all 11 modules.	session with minimum 22 write operations across 11 modules. Audit log accessible only by Hospital Admin role — confirmed by testing all 6 other roles.	logged in Audit Log Coverage Checklist.
QM-09	JWT Security Posture	Three sub-checks: (A) % of API endpoints returning HTTP 401 for expired access token; (B) % of API endpoints returning HTTP 401 for absent token; (C) refresh token endpoint returning HTTP 200 and valid access token within 2 seconds for a valid refresh token.	Sub-check A: 100% of API endpoints return HTTP 401 for expired token. Sub-check B: 100% of endpoints return HTTP 401 with no Authorization header. Sub-check C: Refresh endpoint returns HTTP 200 with a new access token in under 2 seconds in 5/5 consecutive test calls. All three sub-checks must pass before UAT entry is permitted.	Jhon Lloyd Nicolas runs automated test script against DRF router URL list. Refresh timing measured with timestamp logging. Results documented per endpoint.
QM-10	Sandbox-to-Production Parity	Number of defects found in Production attributable to a configuration or environment difference between Sandbox and Production, identified during the Production	0 environment-specific defects identified during Production VM deployment validation. Parity check covers 5 areas: Docker Compose service versions, environment variables,	Mariyah Chavez produces a Deployment Parity Check Report comparing Sandbox and Production across the 5 check areas before handoff. PM reviews and approves the report before Anna Katrina Yturralde is notified for handoff.



		VM deployment validation phase.	PostgreSQL schema version, Cloudflare Tunnel routing, and Fail2Ban configuration — all 5 check areas must match.	
QM-11	Turnover Manual Completeness	For each of the 3 manuals (Training, End-User, Technical): percentage of sections reviewed by PM against the live Production system and confirmed accurate.	100% of sections in all 3 manuals confirmed accurate by PM before submission to WAH. Review completed no later than 5 business days before the scheduled system turnover date.	PM (Elijah Quibin) executes each workflow described in each manual against the live system and marks each section as Confirmed or Needs Revision in the Documentation Review Checklist. Manual submitted only after all sections are Confirmed.
QM-12	WAH Documentation Acceptance Rate	Percentage of submitted manual sections accepted by the WAH reviewer (Anna Katrina Yturalde for Training and End-User manuals; Sir Robert Michael Martinez for Technical Manual) on first submission without requiring revision.	>=90% of submitted sections accepted by WAH reviewer without revision on first submission. If acceptance rate falls below 90%, PM schedules a 30-minute review alignment meeting with Anna Katrina Yturalde/Sir Robert Michael Martinez within 3 business days.	PM tracks WAH reviewer feedback per section in the Documentation Review Checklist. Acceptance rate computed per manual and overall after each WAH review cycle.
QM-13	WAH4P/WAH4PC Sync Accuracy	Percentage of defined WAH4P and WAH4PC sync event types for which 10 consecutive sync	100% of defined sync event types achieve 0 mismatches across 10 consecutive test events per type	Jhon Lloyd Nicolas and Sir Robert Michael Martinez jointly execute sync tests in the integration environment. Results logged in the



		test events produce 0 field-level mismatches between WAH4H source data and WAH4P/WAH4PC target data.	within 30 seconds of the triggering action. Baseline cannot be measured until Sir Robert Michael Martinez provides the interop technical specification — this metric is on hold until that specification is received.	Interoperability Test Report per event type.
QM-14	Milestone Schedule Adherence	For each of the 8 project milestones defined in the Charter, the number of calendar days between the confirmed target date and the actual completion date.	0 milestone completions exceed the confirmed target date by more than 7 calendar days individually. Overall average deviation across all completed milestones ≤ 3 calendar days. Current turnover milestone (May-June 2026) is flagged as At Risk — PM must produce confirmed revised dates and communicate any delay to Sir Pey Canlas no later than 5 business days before the original target date.	PM updates the Milestone Tracking Record after each milestone completion with confirmed date, actual date, and day variance. Variance reported at each sprint review.

3.5.6.2 Metric Escalation Thresholds

Trigger Condition	Escalation Action	Time Window	Escalation Owner
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QM-04: Any Critical defect open at UAT sign-off request	PM cancels sign-off meeting with Anna Katrina Yturalde immediately. Jhon Lloyd Nicolas assigned fix within 1 business day per Critical defect. Re-UAT session scheduled with Anna Katrina Yturalde within 3 business days of fix confirmation. Sir Pey Canlas informed if May-June deadline is impacted.	Cancellation within same business day sign-off meeting was scheduled.	Elijah Quibin (PM)
QM-03: Any module UAT first-pass rate < 90%	Module returned to defect resolution. Jhon Lloyd Nicolas identifies root cause of failures within 1 business day. Re-UAT session for the failed module scheduled with Anna Katrina Yturalde within 5 business days. PM assesses and communicates timeline impact to Sir Pey Canlas if cumulative delays exceed 7 calendar days total.	Root cause within 1 business day of UAT session close.	Elijah Quibin (PM) / Jhon Lloyd Nicolas
QM-07: RBAC Test Matrix < 77/77 combinations at UAT entry	UAT entry blocked immediately. Jhon Lloyd Nicolas and Mariyah Chavez identify untested combinations within 1 business day. All 77 combinations completed and verified before UAT session is rescheduled.	Block applied same day matrix incompleteness is identified.	Jhon Lloyd Nicolas
QM-08: Audit log coverage < 100% of write operation types	Jhon Lloyd Nicolas identifies uncovered write operations within 1 business day of coverage test. Django signal or middleware audit trigger added and deployed to Sandbox. Re-coverage test run within 2 business days. UAT entry blocked until 100% confirmed.	Coverage test re-run within 2 business days of identification.	Jhon Lloyd Nicolas



QM-09: Any API endpoint fails JWT security sub-check	UAT entry blocked. Jhon Lloyd Nicolas fixes affected endpoint permission class within 1 business day. Full JWT test suite re-run against all endpoints. Sign-off not requested until all sub-checks pass.	Fix within 1 business day per failed endpoint.	Jhon Lloyd Nicolas
QM-10: Any environment-specific defect found at Production validation	Production VM deployment halted. Mariyah Chavez validates the Sandbox-Production delta within 4 hours. Jhon Lloyd Nicolas applies technical fixes as needed, then the parity check is re-run before deployment resumes. PM and Sir Robert Michael Martinez notified within same business day.	Delta identified within 4 hours of defect discovery.	Mariyah Chavez / PM
QM-06: Defect resolution velocity < 85% in any sprint	PM meets with Jhon Lloyd Nicolas within 1 business day of sprint close. Sprint carry-over defects triaged and assigned updated resolution dates. Corrective action documented in Risk Register and communicated to Sir Pey Canlas if May-June target is impacted.	Meeting within 1 business day of sprint review.	Elijah Quibin (PM)
QM-14: Any milestone exceeds confirmed date by more than 7 calendar days	PM produces revised milestone schedule within 2 business days of the missed date. Revised schedule communicated to Sir Pey Canlas and Sir Robert Michael Martinez in writing (email) within 2 business days. Risk Register updated with new schedule risk entry and mitigation actions.	Written communication within 2 business days of missed target.	Elijah Quibin (PM)
Any 2 consecutive sprints show the same metric worsening	PM convenes quality review meeting within 3 business days with Jhon Lloyd Nicolas,	Meeting within 3 business days of second	Elijah Quibin (PM)



	Mariyah Chavez, and Kenneth Jajurie. Root cause documented. Corrective action plan with named actions, owners, and completion dates produced. Plan reviewed at next sprint review.	consecutive miss.	
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QUALITY ROLES AND RESPONSIBILITIES (RACI)

The RACI matrix below consolidates all quality-related responsibilities across Team Pseudoers and WAH stakeholders. R = Responsible (does the work), A = Accountable (owns the outcome), C = Consulted (input sought), I = Informed (kept up to date). Each quality activity has exactly one Accountable party.

Quality Activity	PM (Elijah)	Lead Dev (Jhon Lloyd)	QA (Mariyah)	Scrum Master / Agile Dev	Sir Robert Michael Martinez	Ms. Anna Katrina Yturralde	Sir Pey Canlas
FHIR Schema Review (per module)	I	R/A	C	I	C	-	-
Reference Table Audit	I	R/A	C	-	-	-	-
Code Review / PR Gate	I	R/A	C	-	-	-	-
RBAC Matrix Validation	A	R	R	-	C	-	-
Pipeline Gate Validation (L→S→P)	A	C	R	-	-	-	-
Security Testing (JWT, Audit, Encryption)	A	R	R	-	C	-	-
Interop Spec Review (WAH4P/4PC)	A	R	I	-	R/C	-	-
UAT Test Case Preparation	R/A	C	C	I	I	C	-
UAT Execution	A	C	C	I	C	R	-



UAT Defect Logging	A	R	C	-	I	C	-
Post-UAT Defect Resolution	A	R	C	-	C	I	-
Defect Re-Testing	A	R	C	-	-	C	-
Documentation QC (Training/EU/Tech)	R/A	C	I	-	C	C	-
Deployment Parity Check	A	C	R	-	C	-	-
Production VM Deployment Validation	A	C	R	-	R	-	-
UAT Sign-Off	I	I	I	-	C	R/A	-
Final Project Sign-Off	I	I	I	-	C	C	R/A
Quality Metric Reporting	R/A	C	C	C	I	I	I

QUALITY CHANGE MANAGEMENT

3.5.8.1 Change Impact on Quality

Any approved change to project scope, module functionality, or technical standards has the potential to affect established quality baselines. Given that WAH4H is in its post-UAT and pre-turnover phase, changes at this stage carry heightened risk — a change to a module already covered by initial UAT feedback may invalidate prior test evidence and require re-testing before final UAT/sign-off.

All changes during the post-UAT phase must be evaluated for quality impact before approval. The PM is responsible for initiating this evaluation. Changes that affect any module already covered by initial UAT feedback trigger a mandatory re-test of the affected module before the change is included in the turnover package.

Change Type	Quality Impact Assessment	Action Required
Scope addition (new feature or module)	Full QA and QC cycle required for added scope. If added after UAT, entire module must be UAT-tested separately.	PM approval required. Scope added to WBS. QA activities and UAT test cases produced. Anna Katrina Yturralde informed of scope expansion.
Post-UAT defect fix	Fix must be re-tested against the original UAT test case. Adjacent	Fix logged in Defect Log. Re-test confirmed. Re-test result recorded. PM reviews before closure.



	functionality regression-tested if fix touches shared components.	
Standards change (e.g., updated PH Core profile)	Schema, API, and FHIR resource outputs re-validated. Affected modules re-tested. Compliance Matrix updated.	Lead Developer assesses impact. Sir Robert Michael Martinez and PM informed. Affected modules re-enter QC cycle.
Infrastructure change (server config, Docker, Cloudflare)	Deployment parity check re-run. Security configuration re-validated. Pipeline re-tested.	Mariyah Chavez leads re-validation. Production VM deployment blocked until new parity check passes.
Documentation revision post-review	Revised sections re-reviewed by PM before re-submission. If revision affects Technical Manual, Sir Robert Michael Martinez confirms accuracy.	PM signs off on revised sections. Re-submission tracked in deliverables log.

3.5.8.2 Continuous Improvement

Although WAH4H is a project with a defined closure date (May–June 2026), continuous improvement practices are applied within the project lifecycle to progressively improve quality output and team process efficiency. Improvement inputs are gathered from three sources:

- Sprint Retrospectives: Facilitated by John Kenneth L. Jajurie (Scrum Master / Agile Developer). Quality impediments and process gaps identified each sprint are converted into backlog improvement items.
- UAT Feedback: Ms. Anna Katrina Yturralde's UAT feedback and defect observations are reviewed not only for defect resolution but for process root cause — identifying which development or QA process allowed the defect to reach UAT.
- Metric Trends: QM-14 (Schedule Adherence) and QM-06 (Defect Resolution Rate) trends reviewed across sprints. Worsening trends trigger a process review.

Improvement actions are documented in the Lessons Learned Document (WBS 1.11.4) and included in the Project Closeout Report. These serve as the primary quality knowledge transfer artifact from Team Pseudoers to WAH for future WAH4H development phases.

3.5.8.3 Outstanding Quality Risks

The following items from the Compiled Interview Notes (Sir Pey Canlas and Sir Robert Michael Martinez) and the Project Charter represent active quality risks as of May 2026:

Risk	Quality Impact	Status / Action
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May-June turnover deadline at risk (Charter Section 12, Risk 1)	Compressed timeline increases probability of documentation gaps and insufficient re-testing after post-UAT fixes.	High priority. Internal team realignment required. PM must set confirmed milestone dates and communicate any delay to Sir Pey Canlas. (Interview Notes, Section 9, Item 4)
WAH4P/WAH4PC interoperability pending WAH technical specification (R-12 UAT Defects (HIGH/12))	Integration cannot be built, tested, or QC-validated without the technical specification from Sir Robert Michael Martinez. Absence at integration sprint start is a blocking issue.	High priority. PM to escalate to Sir Robert Michael Martinez immediately. Sync events, data fields, and triggers must be defined before integration sprint. (Interview Notes, Section 9, Item 2)
UAT documentation not finalized (Interview Notes, Section 9, Item 7)	Incomplete UAT documentation is a non-negotiable turnover blocker per Sir Pey Canlas (Interview Notes, Section 3). Missing docs prevent formal closure.	Medium priority. Assigned to Jhon Lloyd Nicolas. Must be completed before documentation submission milestone.
Contractual compliance standards not formally confirmed (Interview Notes, Section 9, Item 1)	If contractual standards differ from those currently implemented, additional development and re-testing may be required — directly impacting the at-risk timeline.	High priority. PM to confirm with Sir Pey Canlas at next meeting. All standards in this plan treated as required until confirmation received.
Billing module PHCore alignment gap (known from schema analysis)	Billing module was built FHIR-only without PHCore profile alignment — highest compliance risk module. Compliance gap discovered late requires schema migration and re-testing.	Lead Developer to prioritize Billing module FHIR compliance validation. PHCore alignment must be verified before UAT documentation is finalized.

Approved by the Project Sponsor:

Sir Pey Canlas
Executive Director, Wireless Access for Health (WAH)

Date: _____